

## Introduction

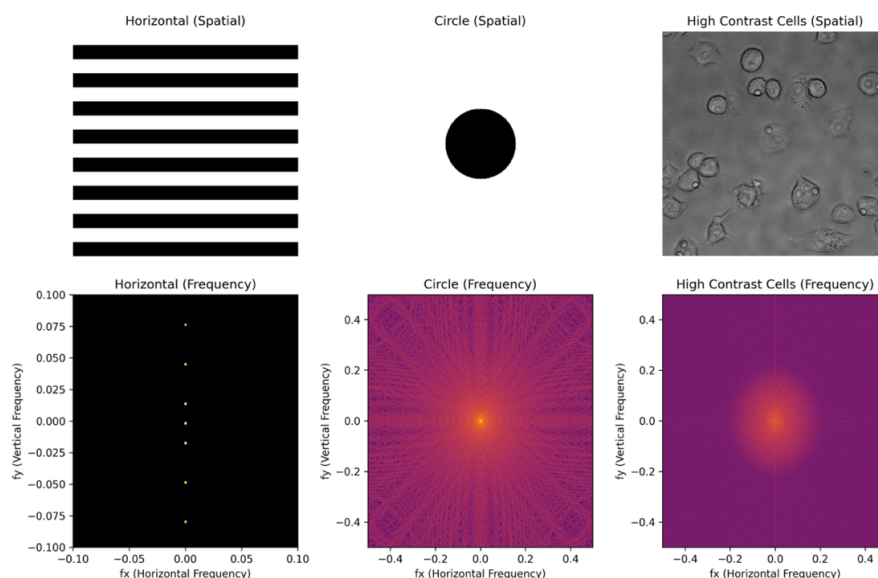
In many biomedical labs, imaging is a primary tool for studying cells, whether to observe structures, track changes, or identify abnormalities. These datasets are often large and complex, making it difficult to extract meaningful information by inspection alone. As a result, image processing plays a key role in interpretation, particularly through tasks such as cell segmentation, which is essential for applications like cell counting in experimental assays.

In image processing, images are two-dimensional signals that can be analyzed in both the spatial and frequency domains. The spatial domain represents pixel intensity directly, while the frequency domain highlights patterns like edges and repeating structures. Convolution is another key operation in the spatial domain, where a small matrix, or kernel, is applied across an image to compute weighted sums of neighboring pixels, allowing local features to be emphasized. While these representations describe the same image, they emphasize different information.

To build intuition, I first analyzed simple patterns in both domains to understand how spatial features map to frequency behavior, later extending this to biological cell images. Building on this, I developed three image-processing pipelines and compared frequency domain enhancement, convolution based segmentation, and a combined approach that uses enhancement to improve segmentation. Together, these methods provide a framework for extracting structure from complex cell images and connecting visual features to their underlying signal representations. This understanding is important for modern image analysis approaches, whether implemented through explicit computational pipelines or data-driven models such as machine learning, all of which rely on effective feature extraction.

## Building Frequency Domain Intuition

The frequency spectra in Figure 1 summarize how representations evolve from simple, structured patterns to more complex distributions. This progression provides the foundation to apply frequency-based techniques to emphasize important features before segmentation.



**Figure 1.** Spatial and frequency representations of simple patterns and a cell image (Note: the horizontal frequency spectra is zoomed in to visualize the discrete points).

To understand this progression, I began by comparing directional patterns. I extended my knowledge of 1D signals to 2D by considering stripes as a two-dimensional square wave. For striped images, the frequency components appeared in the direction of intensity change; the spectrum was oriented perpendicular to the stripes. As shown in Figure 1, horizontal stripes produce components along the vertical axis, composed of discrete points representing the fundamental frequency and higher harmonics. A checkerboard pattern, which varies in both directions, produces components along both axes, forming a 2D harmonic structure that extends my 1D intuition. Similarly, sinusoidal patterns, which vary smoothly between intensity extremes, appear as isolated points in the frequency domain. These behaviors isolate simpler frequency components that compose more complex images.

After considering axial variations, I explored circular shapes as an intermediate step toward biological images. Circles provide a simple model of the roughly round structures seen in cells and introduce radial symmetry. This symmetry appears in the frequency domain as a distribution centered at the origin that spreads outwards rather than aligning along specific axes. As shown in Figure 1, the bright central region with decreasing intensity indicates most information is contained in low-frequency components, while the sharp edges of the circle introduce higher-frequency content. When arranged in a grid, circles introduce periodic structure, producing regularly spaced frequency components similar to the checkerboard pattern.

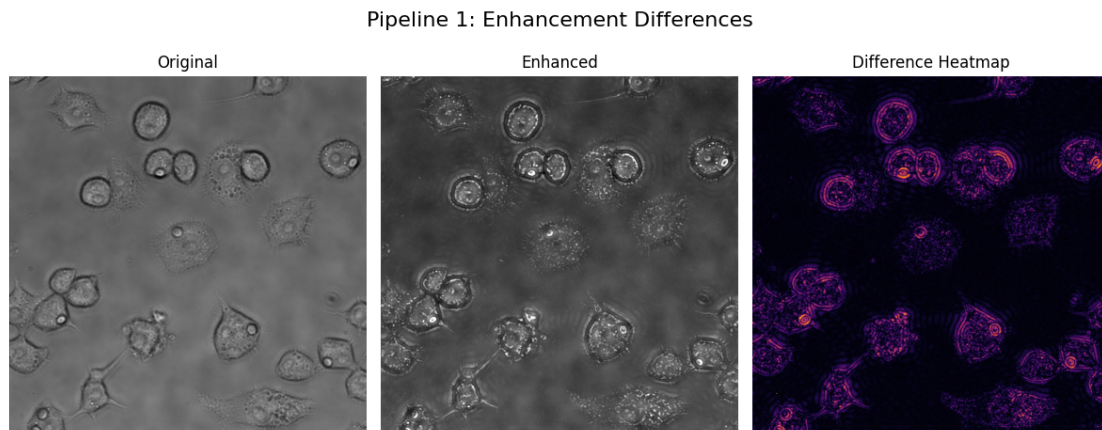
Once I established an understanding of how basic patterns translate into the frequency domain, I explored more complex biological images. Unlike simple patterns, these images lack clear periodicity or symmetry, but their spectra still show strong concentration near the center, indicating dominance of low-frequency information. High contrast images exhibit a broader spread of frequencies, corresponding to sharper edges and more defined structures.

### **Pipeline 1: Frequency Domain Enhancement**

Building on this intuition, I implemented a frequency enhancement pipeline to emphasize structural features in cell images. The goal was to amplify cell boundaries and other rapidly-changing details that correspond to higher frequencies. I transformed the original image using a 2D Fourier transform, and applied a high-pass filter in the frequency domain. This filter acted as a radial mask that removed low-frequency components near the center of the spectrum, which primarily represent smooth background intensity and overall image structure.

After filtering, I transformed the image back to the spatial domain to obtain a version containing primarily high-frequency information. Rather than using this alone, I combined it with the original image. This enhanced the edge features while preserving the overall structure and effectively boosted contrast in regions with rapid intensity changes (without distorting the original image).

As shown in Figure 2, the enhanced image exhibits more pronounced differentiation between the cells and their surroundings. The difference heatmap further highlights where the change occurs, emphasizing the difference in edge and internal detail regions. This confirms this enhancement selectively amplifies high-frequency components.



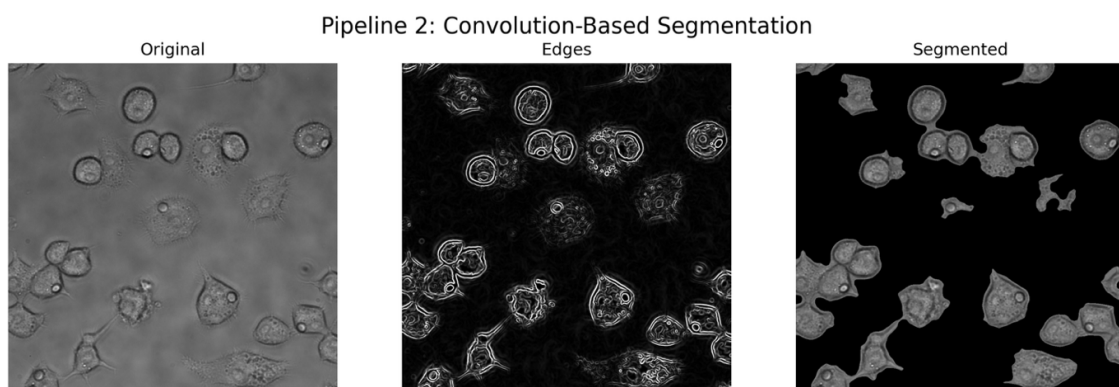
**Figure 2.** Pipeline 1 (frequency domain enhancement): original, enhanced image, and difference heatmap.

### Pipeline 2: Convolution Based Segmentation

After this exploration in the frequency domain, I shifted to convolution in the spatial domain. Convolution naturally extends from one-dimensional signals to two-dimensional images by applying a kernel across both the horizontal and vertical directions. Instead of sliding along a line, the kernel is applied across the image as a matrix. This allows local image structure to be analyzed, with the specific response determined by the kernel values.

I compared a range of kernels to understand how each responded to the images. Edge-detection kernels produced strong response along cell boundaries, while smoothing kernels reduced noise but also blurred important details. Although this approach operates in the spatial domain, it emphasizes the same features I was targeting in the frequency domain filtering. This connection reflects a key signal-processing principle: convolution in the spatial domain corresponds to multiplication in the frequency domain, meaning a kernel effectively acts as a frequency filter that emphasizes specific components of the image. Rapid intensity changes in an image correspond to high-frequency components, and edge-detection kernels highlight these regions by computing differences, or gradients, between neighboring pixel values.

Based on this exploration, I selected an edge-focused approach to generate an edge map, which was then thresholded to produce a binary mask for segmentation. As shown in Figure 3, this method isolates many of the cell regions, although some noise and incomplete boundaries remain.



**Figure 3.** Pipeline 2 (convolution based segmentation): original, edge map, and segmented output.

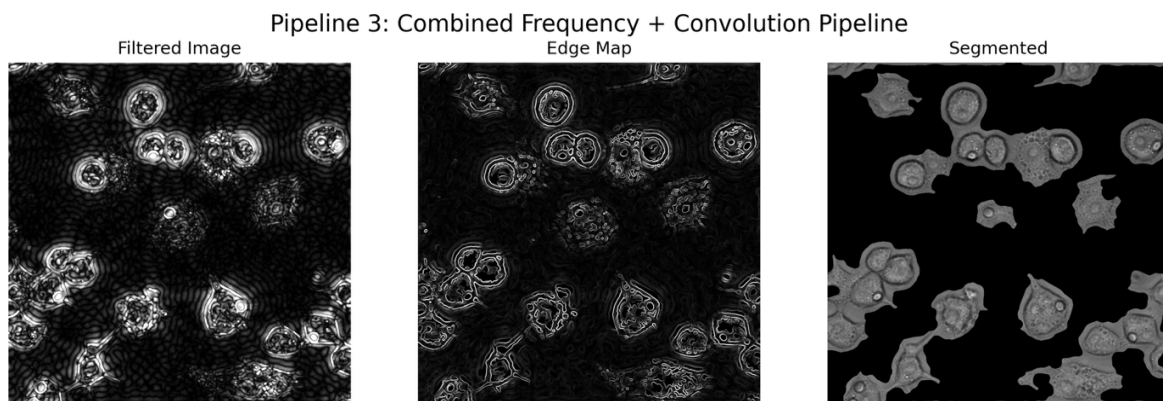
### Pipeline 3: Combined Frequency Enhancement & Convolution Segmentation

After implementing the frequency enhancement and convolution segmentation independently, I wanted to see if combining these approaches could optimize overall segmentation performance. I combined these approaches into a single pipeline that used frequency domain filtering as a preprocessing step to enhance structural features, then applied convolution to more effectively detect and segment these features.

In this pipeline, the image was first enhanced using the frequency domain method developed in Pipeline 1, which amplified high frequency components associated with edges and fine details. As shown in Figure 4, this preprocessing step increased contrast along cell boundaries, making them more distinguishable from the background. The enhanced image was then passed into the convolution based segmentation pipeline, where an edge-detection kernel was applied to generate an edge map. This edge map was subsequently thresholded to produce a binary mask, which was then used to segment the original image.

Compared to Pipeline 2, this approach captures fainter cell details and produces stronger edge responses in low-contrast regions. However, the final segmented outputs remain visually similar. While the combined pipeline improves intermediate representations, this does not always translate to a significantly different final segmentation. When applied to lower-contrast images of cells, the enhanced input produced a more pronounced and continuous edge map, but the final result only varied slightly; convolution alone is already effective at capturing the dominant cell boundaries.

This suggests that frequency domain enhancement and convolution emphasize similar structures in the image through different mechanisms, reflecting how spatial convolution and frequency filtering act as complementary approaches for isolating high-frequency features. Frequency filtering amplifies these features globally, while convolution detects them locally. As a result, combining the two primarily strengthens existing edge information rather than introducing entirely new features, leading to modest differences in the final segmentation.



**Figure 4.** Pipeline 3 (combined frequency + convolution): enhanced input, edge map, and segmentation result.

### Discussion

In this project, I developed and compared three image-processing pipelines to extract structure from biological cell images. Across all approaches, both frequency domain filtering and convolution emphasize regions of rapid intensity change, corresponding to cell boundaries, but they do so differently. Frequency domain enhancement modifies the global

distribution of image information, while convolution detects these features locally through kernel-based operations.

The comparison shows that convolution is often sufficient for capturing the dominant structural features in the image, as seen in Pipeline 2. While frequency domain enhancement improves visibility of fainter details and produces stronger edge responses, this does not substantially change the final segmentation. This indicates that the primary structural information needed for segmentation is already captured by local convolution operations, and that global enhancement mainly reinforces, rather than expands, this information.

A limitation of this approach is its dependence on image quality and parameter selection, such as filter radius, kernel choice, and thresholding. These choices directly affect edge detection and segmentation accuracy and may not generalize across different datasets. This limitation becomes more significant in complex biological images, such as dense tissue or brain cells, where irregular shapes and closely packed structures can lead to overlapping regions in the segmentation mask. In these cases, convolution based methods may struggle to fully separate adjacent cells, particularly when boundaries are not well defined. This is similar to processing noisy one-dimensional signals, where additional or overlapping information can obscure the underlying signal, making it difficult to discern which features are meaningful. As a result, while this approach works well for simpler, well-separated cell structures, additional processing or refinement may be helpful for more complex or noisy images.

## **Conclusion**

Initially, I expected that combining these approaches would optimize cell identification and segmentation, but I found that once key features like edges are already captured, additional processing often reinforces existing information rather than introducing new structures. Overall, this project demonstrates how frequency domain analysis and convolution can be used together to interpret and extract structure from biological cell images. By building intuition from simple patterns and extending it to real cell data, I showed how spatial and frequency representations highlight different aspects of the same image. The results indicate that convolution based methods are effective for directly capturing and segmenting dominant cell boundaries, while frequency domain enhancement improves the visibility of finer details and low-contrast features. While I didn't create a novel segmentation approach, it was interesting to see how these two methods could target the same details in parallel and complement each other.

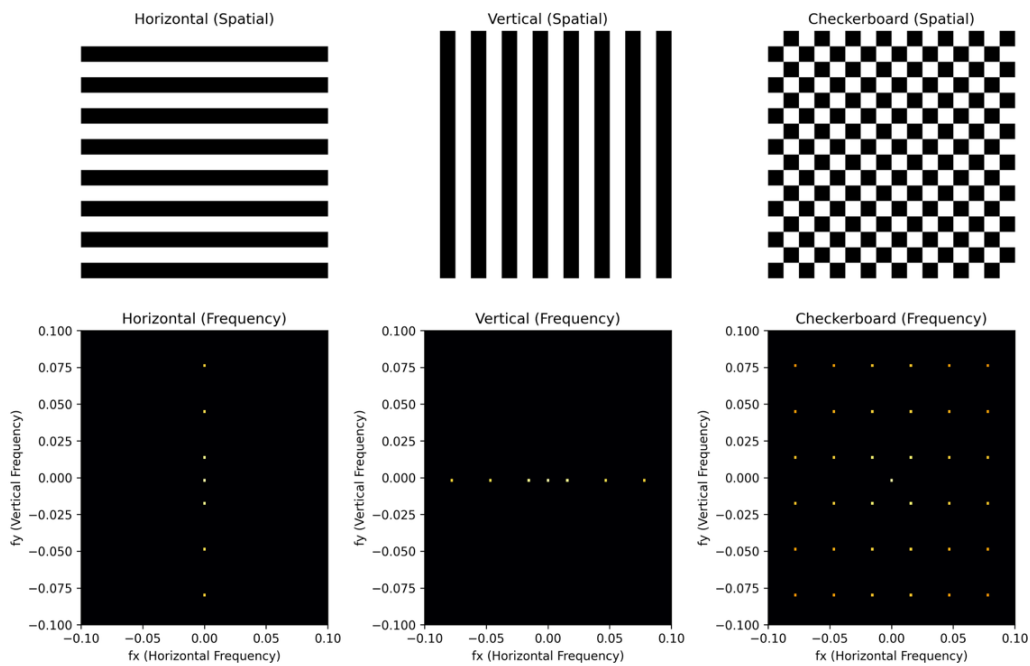
Ultimately, I realized that effective image analysis depends less on adding complexity and more on understanding how different transformations target meaningful features. My explorations showed that highpass filtering extracts regions associated with rapid changes in pixel intensity by isolating high-frequency components, while convolution kernels detect these same features locally by computing intensity gradients across neighboring pixels. These findings are especially important in biomedical imaging, where analyzing large datasets is crucial but variability in image quality, structure, and noise can impact results. Effective image analysis is not about applying more methods, but understanding how features are represented in the signal and choosing the transformations that isolate those features effectively.

## Acknowledgements

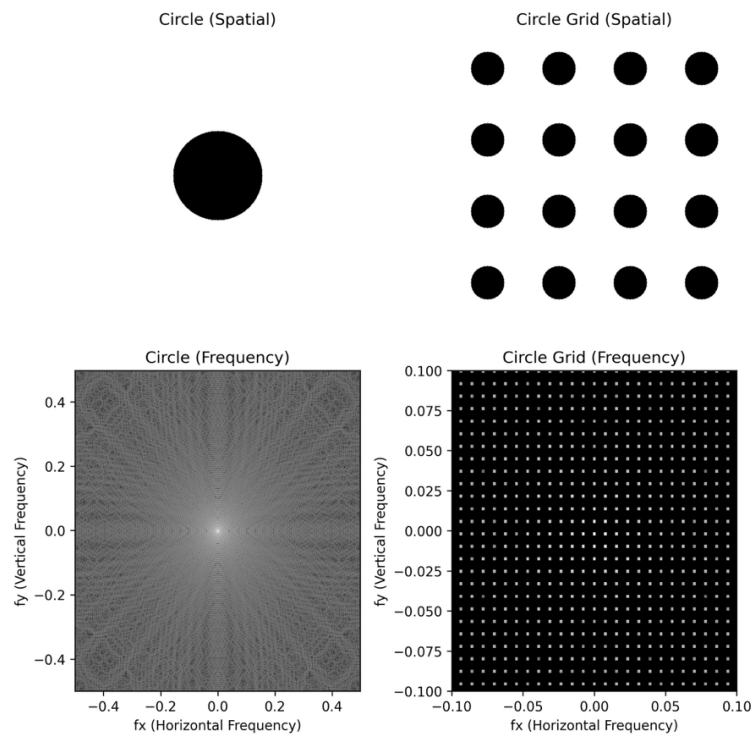
1. Additional figures are included below to provide further context for my explorations.
2. My coding repository for this project can be found here:
  - a. <https://gitlab.oit.duke.edu/mcs131/bme271-finalproject-soto>
3. I referred to some sources for images and to build background:
  - <https://www.cellimagelibrary.org/images/27219>
    - Low contrast cell image
  - <https://pubs.acs.org/doi/10.1021/acsami.5c25569>
    - I contributed to work related to this segmentation pipeline and used similar methods to guide my convolution approach, as well as the high-contrast cell image.
  - <https://opencv.org/edge-detection-using-opencv/>
    - Reference on Open CV edge detection
  - <https://thepythoncodingbook.com/2021/08/30/2d-fourier-transform-in-python-and-fourier-synthesis-of-images/>
    - Interesting source on 2D Fourier Transform in Python and Sinusoidal Gratings
  - [https://medium.com/@timothy\\_terati/image-convolution-filtering-a54dce7c786b](https://medium.com/@timothy_terati/image-convolution-filtering-a54dce7c786b)
    - Good explanations / visuals of 2D convolution

Thank you for a very engaging and intellectually stimulating semester – I really enjoyed this course!

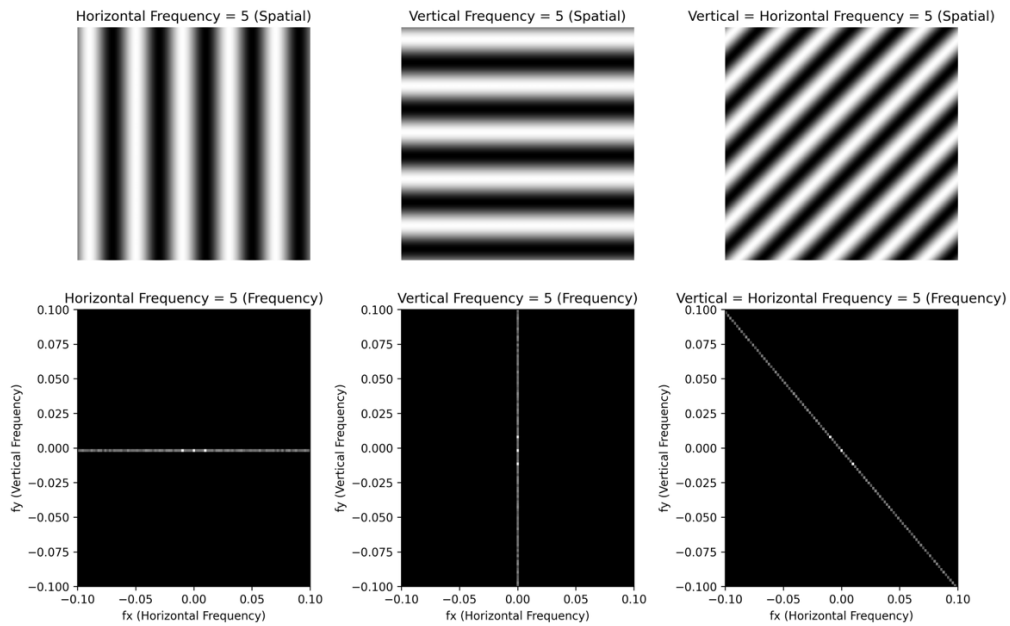
## Appendix - Additional Figures



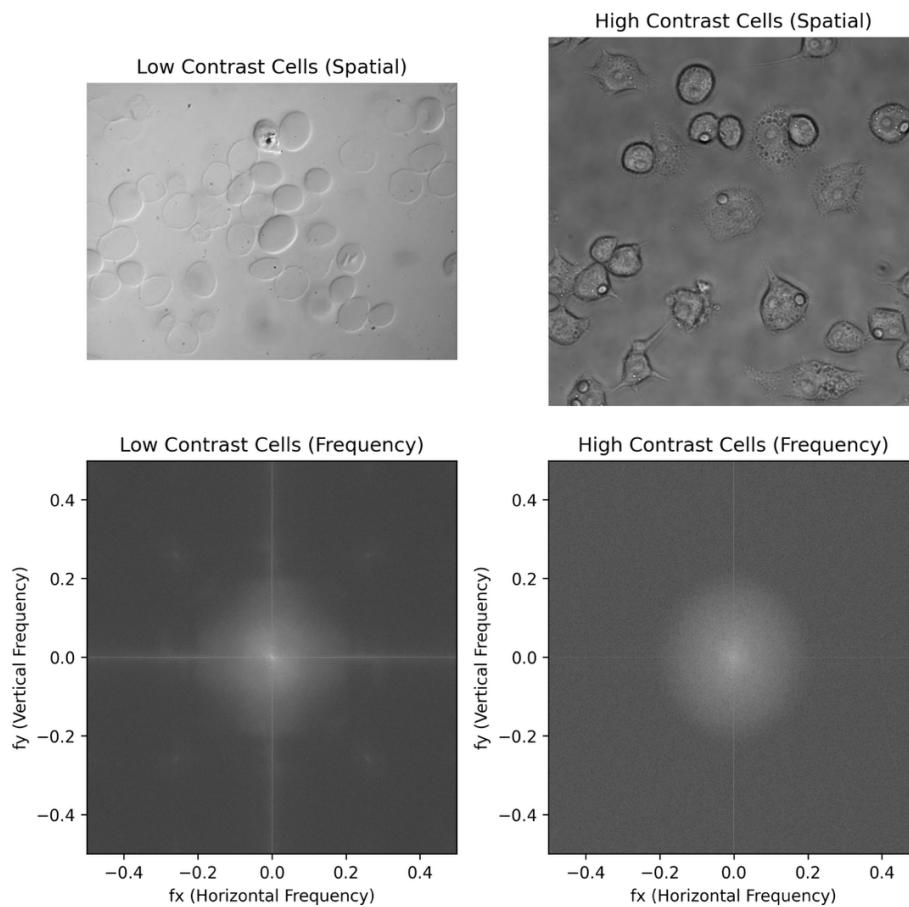
**Figure 5.1.** Spatial (top) and frequency (bottom) representations of horizontal, vertical, and checkerboard patterns. Frequency components appear perpendicular to stripe direction, while the checkerboard produces a grid of discrete frequencies due to variation in both directions.



**Figure 5.2.** Spatial (top) and frequency (bottom) representations of a circle and a circle grid. The single circle produces a radially symmetric frequency distribution, while the grid introduces regularly spaced frequency components due to spatial repetition.

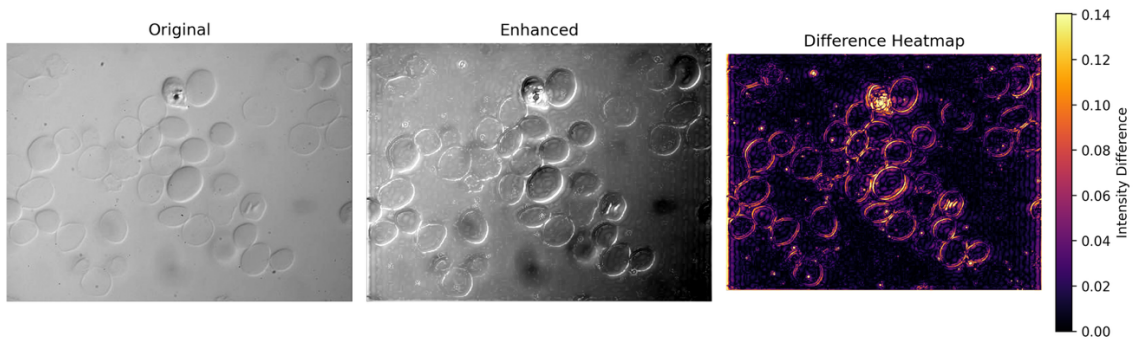


**Figure 5.3.** Spatial (top) and frequency (bottom) representations of sinusoidal patterns with horizontal, vertical, and diagonal orientations; each sinusoid produces frequency components perpendicular to its direction of variation.

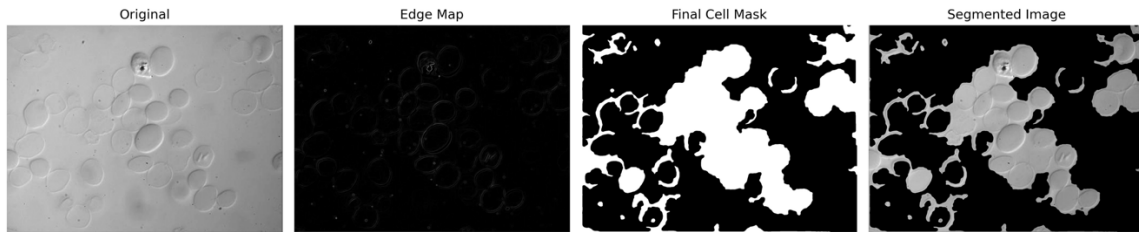


**Figure 5.4.** Spatial (top) and frequency (bottom) representations of low- and high-contrast cell images. Both are dominated by low-frequency content, but the high-contrast image shows a broader frequency spread, reflecting sharper edges and increased structural detail.

### Pipeline 1: Frequency-Domain Enhancement Differences

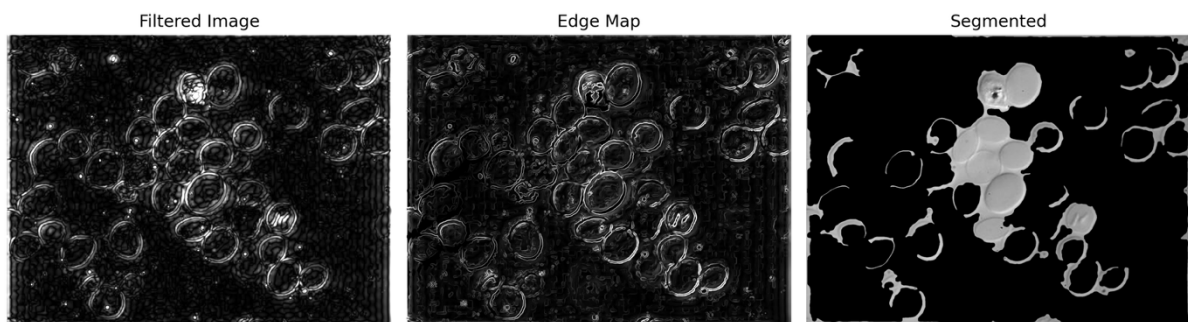


**Figure 6.** Frequency domain enhancement results showing the original image, enhanced image, and absolute difference heatmap. High-frequency filtering amplifies edge and boundary features, with the heatmap highlighting regions of greatest intensity change (corresponding to cell structures).

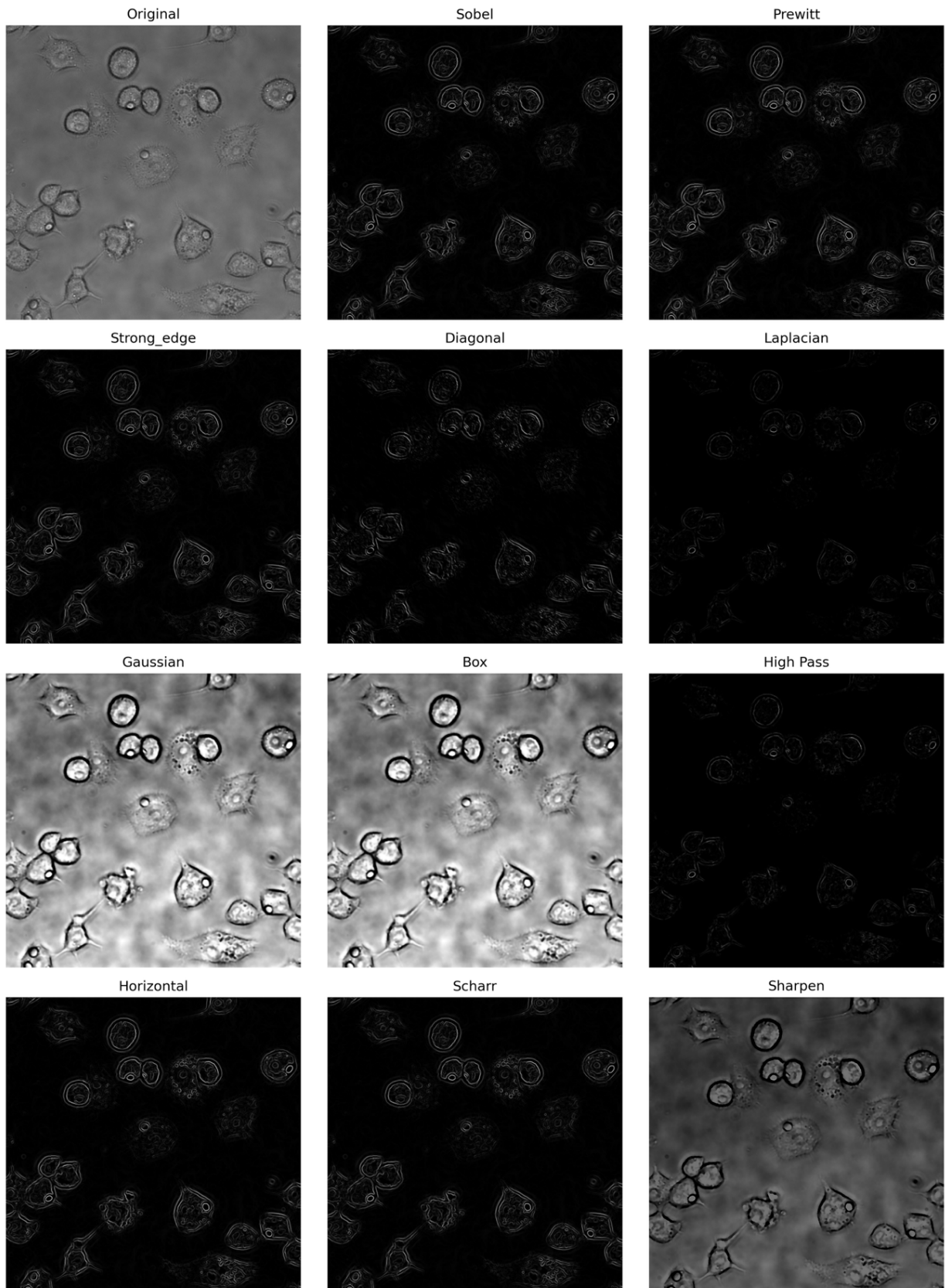


**Figure 7.** Convolution based segmentation of a low-contrast cell image. The edge map highlights intensity transitions, which are thresholded to produce a binary mask, isolating cell regions and enabling extraction of segmented cells from the original image.

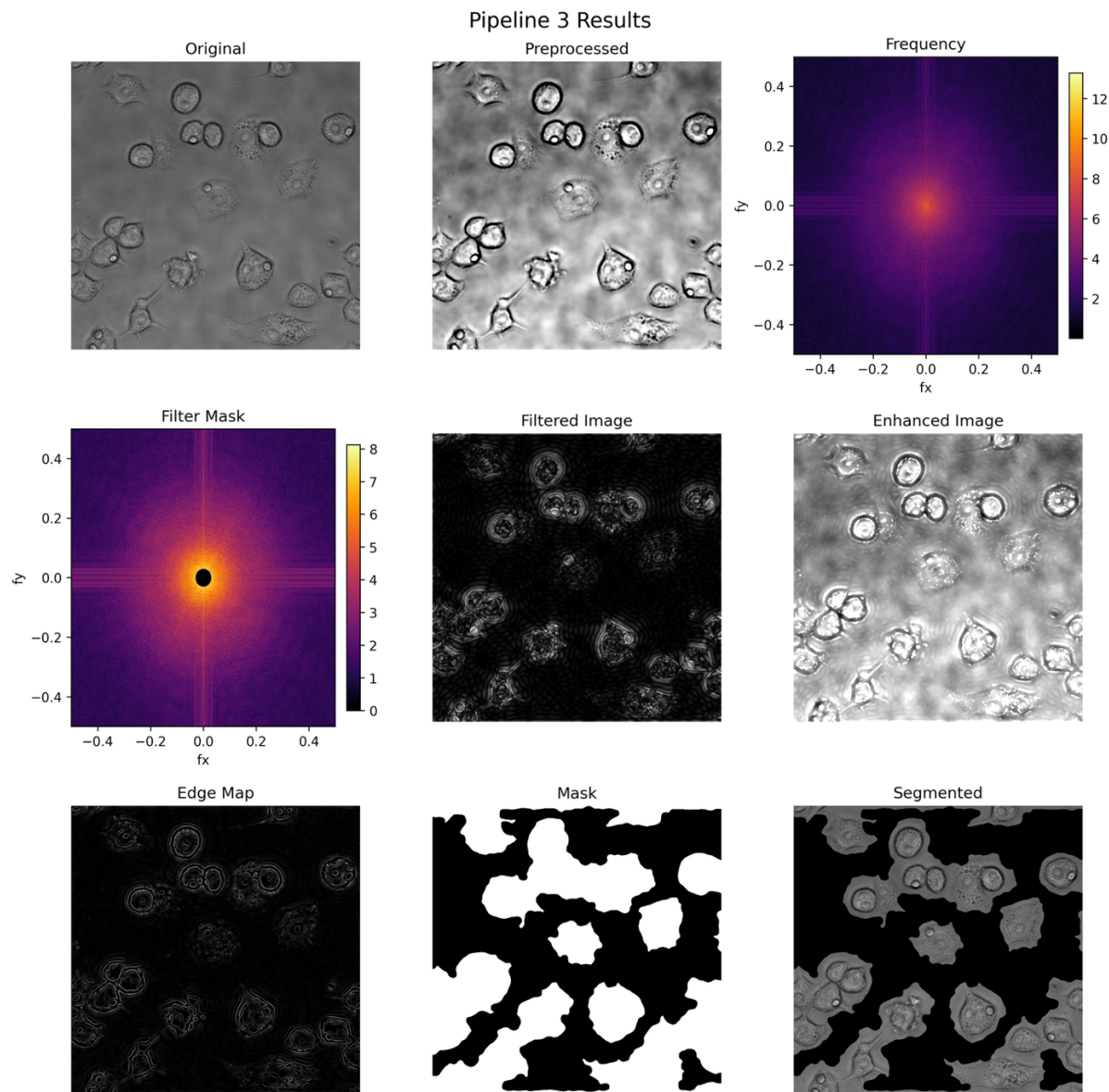
### Pipeline 3: Combined Frequency + Convolution Pipeline



**Figure 8.** Combined frequency and convolution pipeline results showing the filtered image, edge map, and final segmentation. Frequency domain enhancement improves edge visibility before extraction of cell regions through convolution based segmentation.



**Figure 9.** Comparison of convolution kernels applied to a cell image. Edge-detection filters (e.g., Sobel, Prewitt, Scharr) highlight cellular boundaries, while smoothing (Gaussian, box) and sharpening filters alter contrast and detail, illustrating how kernel choice impacts feature extraction.



**Figure 10.** Combined frequency and convolution pipeline for cell segmentation. Frequency domain filtering enhances edges, which are then detected and thresholded to produce a mask and final segmented image.